

OISD-STD-185

Wire Line Operations

Industry: Oil & Natural Gas

Category: Exploration & Production (Well Servicing)

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1. STANDARD OVERVIEW

Standard Number: OISD-STD-185

Official Title: Wire Line Operations

Type: Standard (STD)

Industry Category: Exploration & Production (Upstream)

Main Subject: Establishing mandatory safety guidelines and operational protocols for executing wireline and slickline operations in oil and gas wells. The standard focuses on the mechanical, pressure, and operational hazards associated with lowering and retrieving tools in pressurized wellbores.

2. PURPOSE OF THE STANDARD

Wireline operations involve lowering tools into live (pressurized) oil and gas wells using a continuous spool of wire or cable. Because these operations are often performed on producing wells without "killing" them, there is a constant risk of high-pressure fluid release, blowouts, and severe mechanical injuries from parted wires under high tension.

The purpose of **OISD-STD-185** is to provide a standardized framework for the safe rigging up, operation, and maintenance of wireline equipment (including slickline and braided line). It ensures that well pressure is safely contained at the surface and that the mechanical integrity of the hoisting systems protects the crew.

3. SCOPE

The standard comprehensively covers the safety requirements for:

- **Slickline and Braided Line Operations:** Lowering mechanical tools for well intervention, pulling safety valves, and clearing obstructions.
- **Electric Line (E-Line) Operations:** Although heavily integrated with OISD-183 (Logging Operations), this standard addresses the mechanical rig-up of E-line operations.
- **Pressure Control Equipment (PCE):** Safe installation, testing, and operation of lubricators, stuffing boxes, and wireline blowout preventers (BOPs).
- **Rigging Up and Rigging Down:** Working at heights, scaffolding, crane lifting, and safe securing of the tool string at the wellhead.

- **Wire Maintenance:** Required testing of the wire for ductility and tension before operations.

4. INDUSTRY APPLICATION

OISD-STD-185 is heavily utilized during the **Production and Well Servicing Phases**. Key users include:

- Wireline and Slickline Service Providers executing well interventions.
- Production Seismologists and Reservoir Engineers acquiring downhole data.
- Wellsite Supervisors overseeing safe intervention on live platforms/onshore pads.
- HSE Auditors verifying that rig-up procedures and PCE testing meet statutory safety norms.

5. MAJOR AREAS COVERED

- Site Preparation and Pre-Job Planning
- Rigging up Pressure Control Equipment (PCE)
- Pressure Testing Requirements
- Safe Operation of the Wireline Winch Unit
- Wellhead Valve Management
- Mechanical Wire Integrity and Testing
- Emergency Procedures (Cutting wire, BOP activation)

6. DETAILED TOPIC-WISE SUMMARY

Rigging Up Pressure Control Equipment (PCE)

What it covers: The correct sequence for installing the surface barrier system on top of the Christmas Tree (wellhead).

Important Requirements: Two active barriers between the well pressure and the environment (e.g., swab valve and upper master valve) must be closed during rig-up. Tag lines must be used when lifting heavy lubricators. Personnel working at heights (typically over 6 feet) to assemble the PCE must use appropriate scaffolding and fall protection.

PCE Pressure Testing

What it covers: Verifying that the surface equipment can withstand well pressure before the well is opened.

Important Requirements: A full PCE rig-up pre-job pressure test must be performed every time equipment is rigged up. The lubricator, BOP, and stuffing box must be hydrostatically tested to ensure seals and quick-union O-rings are perfectly intact prior to exposing them to wellbore pressure.

Winch Unit Operations

What it covers: Safety rules for the operator controlling the wireline drum.

Important Requirements: The operator must have a clear line of sight to the lower sheave (hay pulley) and the stuffing box. The winch and the truck must be properly grounded to dissipate static electricity. The tension indicator (weight indicator) must be constantly monitored to avoid pulling the wire beyond its safe working load or parting it.

Wellhead Valve Management

What it covers: Rules for interacting with the well's permanent valves.

Important Requirements: The lower master valve must never be used for routine operations; it is an emergency isolation barrier only. When opening valves to allow tools into the well, operators must slowly equalize pressure across the valves to prevent explosive decompression and seal damage.

7. EQUIPMENT AND SYSTEMS

Equipment / System	Purpose	Standard Context
Wireline BOP	Contains well pressure if the lubricator fails or if the wire needs to be cut/sealed.	Must be correctly sized for the wire being used (e.g., 0.108" or 0.125" slickline rams).
Lubricator	A pressurized pipe section holding the tool string above the wellhead before deployment.	Must be pressure tested before opening the well; never stand under it when suspended.
Stuffing Box	Seals around the moving wire at the very top of the PCE.	Prevents gas/oil from escaping while allowing the wire to spool in and out of the well.

Equipment / System	Purpose	Standard Context
Hay Pulley (Sheave)	Guides the wire from the winch unit horizontally to the wellhead.	Must be secured to the wellhead using certified chains/slides (ropes are forbidden).

8. SAFETY REQUIREMENTS

- **Tension Safety:** The wire must never be pulled beyond its Safe Working Load (SWL) to prevent "parting" (snapping) which can cause the wire to violently whip back and cause fatal injuries.
- **Zone Classification:** Wireline equipment must comply with hazardous area zone rules. In Zone 2 environments (e.g., offshore platforms), electrical systems on the winch unit must be explosion-proof.
- **Hydrogen Sulfide (H₂S):** If operating in an H₂S environment, the PCE metallurgy must comply with sour gas standards (e.g., NACE MR0175), and the crew must have breathing apparatus ready.

9. OPERATIONAL REQUIREMENTS

A minimum crew size (typically three personnel) is mandated to ensure someone is always monitoring the winch, checking the wellhead, and available for emergency response. Night operations are generally discouraged unless adequate and sufficient illumination covering the entire work site is available. Offshore, rigging up on unmanned jackets must cease before dark.

10. INSPECTION, TESTING & MAINTENANCE

Continuous wire integrity testing is a critical component of OISD-185. A "wire history card" must be maintained accurately. Operators must physically test the wire for ductility (e.g., wrap tests or torsion tests) at regular intervals (or at the start of a new operation) and cut off the fatigued sections at the tool attachment point (the rope socket) to prevent downhole failures.

11. EMERGENCY & HSE ASPECTS

The standard defines emergency protocols for "cutting and dropping" the wire if the well begins to blow out and the surface PCE fails. The wireline BOP includes shear rams capable of cutting the wire and

sealing the wellbore. Proper manual handling of the wire is also emphasized; operators must hold the wire near the end when cutting to prevent it from flying loose and causing eye or facial injuries.

12. IMPORTANT TECHNICAL CONCEPTS

Slickline vs. E-Line: Slickline is a single, smooth strand of metal wire used strictly for mechanical work (setting plugs, shifting sliding sleeves). E-Line (electric line) is a braided cable containing a conductive core used for logging and transmitting data.

Pressure Equalization: Before opening the swab valve to drop tools from the lubricator into the well, the pressure inside the lubricator must be brought up to match the wellhead pressure to prevent the tools from being blown violently upward.

Tool String: The heavy assembly of jars, stems, and tools attached to the end of the wire, providing the weight needed to overcome the upward force of well pressure acting against the wire's cross-sectional area.

13. IMPORTANT TERMS & ABBREVIATIONS

Term / Abbreviation	Meaning	Relevance
PCE	Pressure Control Equipment	The entire surface stack (BOP, lubricator, stuffing box).
BOP	Blowout Preventer	The primary safety valve capable of sealing around or cutting the wire.
WHFP / CP	Well Head Flowing Pressure / Casing Pressure	Metrics that must be recorded before operations begin.
SWL	Safe Working Load	The maximum allowable tension applied to the wireline.

14. RELATED STANDARDS & REGULATIONS

- **OISD-STD-183 (Logging Operations):** Highly complementary standard that dictates the handling of explosives and radioactive sources frequently deployed via wireline.

- **OISD-GDN-182 (Workover Operations):** Broad well servicing guidelines under which specific wireline tasks are often executed.
- **API STD 6A / API Spec 16ST:** American Petroleum Institute standards frequently referenced for the technical design and torque specifications of flanged wellhead connections and coiled tubing/wireline equipment.

15. PRACTICAL USE CASES

Scenario 1: Rigging Up on a Live Gas Well

Relevant aspect: Rigging Up PCE & Pressure Testing.

Why it matters: Before opening the well to retrieve a downhole safety valve, the crew follows OISD-185 to install the lubricator and wireline BOP. They hydrostatically test the assembly to the maximum anticipated wellhead pressure to ensure no gas leaks occur when the well is opened.

Scenario 2: Wire Breakage Prevention

Relevant aspect: Inspection, Testing & Maintenance.

Why it matters: To prevent the slickline from snapping deep inside the well, the operator cuts off the first 30 feet of wire, performs a torsion wrap test on the fresh end as per standard procedure, and accurately logs the results in the wire history card before tying on the tool string.

Scenario 3: Emergency Well Closure

Relevant aspect: Emergency Procedures.

Why it matters: If the stuffing box at the top of the lubricator blows out, the operator immediately activates the Wireline BOP rams to seal around the wire, stopping the hydrocarbon leak while preserving the wire intact, preventing an environmental disaster.

16. KEY TAKEAWAYS

1. OISD-STD-185 governs the mechanical and operational safety of slickline and wireline well interventions.
2. Pressure Control Equipment (PCE) must be rigorously pressure-tested prior to every job.
3. Two active pressure barriers must be maintained during rig-up operations.
4. The winch operator must have an unobstructed line of sight to the wellhead and hay pulley.
5. Wireline BOPs must be appropriately sized for the specific diameter of wire being run.
6. The mechanical integrity of the wire must be continuously tracked via a wire history card and periodic physical testing.

7. Operators must never rely on the lower master valve for routine operations; it is strictly an emergency barrier.
8. Special precautions apply for working at heights (scaffolding) and lifting heavy lubricators (tag lines).

17. INFORMATION LIMITATIONS

- **Verified Information:** General scope, PCE testing requirements, basic rig-up safety protocols, winch operation guidelines, and wire testing maintenance records.
- **Information Requiring Official Access:** Specific numeric torque values for wellhead flanges, exact hydrostatic test duration times, specific formulas for calculating SWL limits based on wire metallurgy, and exact zone clearance radii.

Note: "Detailed requirement is not publicly available in the sources reviewed." No quantitative testing durations, metallurgical formulas, or exact torque values have been fabricated.

18. SOURCES & REFERENCES

1. **Oil Industry Safety Directorate (OISD)** - Official Standards List & Publication Records (Confirming OISD-185 Wire Line Operations, Aug 2002).
2. **Scribd Document Repository** - Slickline Operations Safety Practices and Guidelines (Extracts detailing PCE rigging, wellhead valve management, wire testing, and winch operation protocols typical to OISD-185/API standards).
3. **Oil India Limited** - Public Tender Documents (Referencing compliance to OISD-185 and safety precautions for 10,000 PSI wireline BOPs).